



P.C.M. QUOTATION SYSTEM

System Setup & Maintenance Manual

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System Stack: MERN (React, Express, Node.js, MongoDB Atlas)

1. System Overview

The P.C.M. Quotation System is a specialized web application designed for **PCM Cosmetic Products Trading**. It streamlines the generation, customization, and management of business price quotations. The system integrates a modern single-page-application frontend with an autonomous backend server and cloud data persistence.

Component	Technology Stack	Description
Frontend (UI)	React (v19), Vite, Tailwind CSS	Provides a responsive, fast-loading user interface running locally in the browser.
Backend (API)	Node.js, Express, Mongoose	Handles endpoints, business logic validations, and talks to the database.
Database	MongoDB Atlas (Cloud)	Stores products, sizes, prices, and system data. Hosted securely in the cloud.
Execution	Concurrently (npm package)	Launches both frontend and backend servers together with a single command.

2. System Prerequisites

Before installing the system, ensure the following software is installed on the target machine:

- **Node.js (v20.x or higher):** Crucial for running both the frontend and backend JS runtimes. Downloader can be found at <https://nodejs.org/>.
- **Git or GitHub Desktop:** Necessary for developers/maintainers to clone the codebase and manage version releases. Downloader available at <https://desktop.github.com/>.
- **Web Browser:** A modern browser (Google Chrome, Microsoft Edge, or Safari) to display the system interface.
- **Internet Connection:** Required for the first installation (to fetch npm modules) and for database CRUD operations (cloud MongoDB connectivity).

3. Automated Installation

The project provides pre-compiled setup executables (**setup-win.exe** for Windows and **setup-macos** for macOS) to automate the setup process. These executables automatically detect if Node.js is present, download/install it if missing, install all required dependencies recursively, and create desktop launchers.

3.1 Windows Installation Guide

1. Navigate to the project root folder (PCM_QUOTATION).
2. Double-click the file named **setup-win.exe** (which invokes the underlying install.bat installer).
3. **Node.js Check:** The script checks for Node.js. If missing, it will automatically download the installer (.msi), open the wizard, and wait for completion.
4. **Dependency Resolution:** Once Node.js is present, the script executes npm install in the root, backend/, and frontend/ folders.
5. **Shortcut:** A shortcut named **PCM Quotation System** with the custom icon is generated on your Windows Desktop.

3.2 macOS Installation Guide

1. Open the project root folder in Finder.
2. Double-click the file named **setup-macos** (which invokes the underlying install.command installer).
3. **Node.js Check:** If Node.js is missing, the script fetches the official macOS installer package (.pkg) via curl and opens the installation wizard.
4. **Dependencies:** The script installs all npm modules recursively inside root, backend, and frontend environments.
5. **Desktop Shortcut:** A launcher script named **PCM Quotation System.command** is created directly on your Desktop.

Note on Permissions (macOS)

On macOS, the first time you run the setup-macos executable or a launcher script, you may encounter security warnings. To bypass this, right-click the file, select **Open**, and confirm the prompt, or open Terminal and execute:
chmod +x setup-macos install.command startup.command.

4. Manual Installation

If the automated batch or bash scripts are blocked by corporate security policies or if you are setting up a custom environment, you can initialize the system manually using the terminal command line.

1. Open terminal/command prompt and change directory to the project folder:

```
cd /path/to/PCM_QUOTATION
```

2. Install the root workspace package manager dependencies (e.g., concurrently):

```
npm install
```

3. Navigate into the backend directory and install its specific server packages:

```
cd backend  
npm install
```

4. Navigate back and enter the frontend directory to install client dependencies:

```
cd ../frontend  
npm install
```

5. Ensure the backend configuration file exists (see Section 6). Once completed, return to root folder.

5. Running the System

The system runs as two local servers (Vite frontend + Node backend) communicating via HTTP API requests.

5.1 Standard Execution (Recommended)

- **Windows:** Double-click the **PCM Quotation System** shortcut on your Desktop, or run startup.bat in the project folder.
- **macOS:** Double-click the **PCM Quotation System.command** shortcut on your Desktop, or run startup.command in the project folder.

What happens: A terminal window will launch both the backend and frontend simultaneously, wait 4-5 seconds for database connection handshakes, and then automatically launch your default browser redirected to:

```
http://localhost:5173
```

5.2 CLI Execution (Alternative)

You can manually startup the servers using terminal commands directly from the root folder:

```
npm run start
```

This fires the concurrently script that launches: node --watch --env-file .env index.js (backend) and vite (frontend).

6. Backend & Database Configuration

The system stores products, packaging sizes, pricing models, and system parameters in a cloud-hosted **MongoDB Atlas** database. The connection settings are stored locally in the environment configuration.

6.1 Environment Configuration File (.env)

The backend server checks the file: backend/.env on startup. This file must contain the following variables:

```
SERVER_PORT = 6942
MONGODB_URI = "mongodb+srv://cosmeticproductstradingpcm:pcm_admin_2025@pcm-database.ymtfu.mongodb.net/pcm_database?retryWrites=true&w=majority&appName=PCM-Database"
```

6.2 Cloud Database Access Credentials

For system administration, database maintenance, or raw data edits, developers can access the cloud dashboard:

- **MongoDB Console:** <https://cloud.mongodb.com/>
- **Google / MongoDB Login Email:** cosmeticproductstradingpcm@gmail.com
- **Shared Administrative Password:** pcm_admin_2025

The GitHub code repository is linked to the same administrative organization: **PCM Cosmetic Products Trading**.

7. Maintenance & Troubleshooting

Here are resolutions to common issues that might arise during setup or execution:

Issue / Symptom	Probable Cause	Resolution
Node command not found / Node.js missing	Node.js is either not installed or not added to system PATH variables.	Re-run the automated installer or download Node.js installer from nodejs.org. Restart terminal/PC after installation.
Error: Port 5173 or 6942 already in use	A previous instance of the server is still running in the background.	Close all command prompt/terminal windows. On Windows, open Task Manager and end 'Node.js JavaScript Runtime' tasks. On macOS, run killall node.

Database connection timeout (Mongoose error)	No internet connectivity, or Mongo Atlas IP whitelist block, or wrong MONGODB_URI.	1. Check internet connection. 2. Verify backend/.env contains correct URI. 3. Log in to MongoDB Atlas cloud dashboard to ensure the user's IP is allowed by IP Access Rules.
Blank page in browser / Console error 'Network Error'	Backend API server did not launch correctly or crashed, meaning frontend cannot fetch sizes/products.	Verify backend console output for errors. Run <code>npm run backend_start</code> manually in the backend folder to view crash stacks.

Administrative & Support Contact

If you encounter critical bugs, deployment failures, database configuration issues, or questions regarding MongoDB schemas, please contact the system administrator:

Email: ybutgianpaolo@gmail.com